

ZALA Lancet

The **ZALA Lancet** (official designation: *Item 52/Item 51*)^[1] is an unmanned aerial vehicle (UAV) and loitering munition developed by the Russian company ZALA Aero Group (part of Kalashnikov Concern) for the Russian Armed Forces. It was first unveiled in June 2019 at the ARMY-2019 military expo in Moscow.^{[2][3]} It is a further development of the ZALA KYB-UAV (also known as KUB-BLA) loitering munition.^[4]

Production

Russian state channel Rossiya-1 journalists visited a factory, where Lancets are produced. As reported in the video, "the real number of lancets is disclosed, but as the developers assure, for every Western tank, including those which have not even been deployed yet, several drones have already been stocked". It is also said that since the beginning of the Russia's invasion of Ukraine, the production of Lancet drones was increased by a factor of 50.

Description

The ZALA Lancet can be used for both reconnaissance and strike missions. It has a maximum range of 40 kilometres (25 mi) and a maximum takeoff weight (MTOW) of up to 12 kilograms (26 lb). In combat mode, it can be armed either with high explosive (HE) or HE-fragmentation warheads. It features optical-electronic guidance and TV guidance unit, which allows the munition to be controlled during the terminal stage of flight.^[3] The drone features intelligence, navigation and communications modules.^[5] According to Zala Aero Chief Designer Alexander Zakharov, the Lancet can be used in so called "air mining" role. In this role, the drone dives at maximum speed up to 300 kilometres per hour (190 mph) and strikes enemy unmanned combat aerial vehicles (UCAVs) mid-flight. Lancet can be launched via catapult launcher from ground or sea platforms like the Raptor-class patrol boats.^[6] The drone is powered by an electric motor.^[7]

Russian sources state a Lancet costs about 3 million rubles (\$35,000).^[8]

Lancet utilizes the Jetson TX2 module by NVIDIA as its onboard control equipment and the Xilinx Zynq SoC module by Xilinx, an American company owned by AMD, for implementing programmable logic.^[9]

Operational history



ZALA Lancet, front view



ZALA Lancet, side view

The Lancet has been combat tested in Syria during the Russian military intervention in the Syrian civil war since at least November 2020.^[5] In April 2021, it was engaged in strikes against Tahrir al-Sham in the Idlib Governorate.^[6]

On 8 June 2022, Russian defence corporation Rostec announced that Lancet and KUB drones were deployed during the 2022 Russian invasion of Ukraine.^[10] A month later, the first video of their combat use in Ukraine emerged.^[4] In late 2022, multiple videos appeared on social networks showing Lancet drones striking a variety of Ukrainian military targets, such as air defence systems, self-propelled howitzers, tanks and military trucks. Among the targets damaged or destroyed were S-300 missile systems, a Buk-M1 missile system, a T-64 tank, Western-supplied M777 and FH70 howitzers, as well as M109, AHS Krab and CAESAR self-propelled howitzers.^{[11][12][13]} On 4 November 2022, a Gyurza-M-class gunboat of the Ukrainian Navy was damaged by a Lancet drone, the first time a Lancet attacked a naval target during the war.^{[14][12]}

In February 2023, a Lancet drone targeted and struck a rare Ukrainian T-84 tank.^[15] In March 2023, a British-supplied Stormer HVM air-defence system was struck by a Lancet drone for the first time.^[16]

While long-range Iranian Shahed-136 kamikaze drones are used against Ukrainian energy infrastructure, the Lancet is employed as a precision battlefield weapon against high-value military targets, usually located by a reconnaissance drone before it gets launched. Although the Lancet is recorded having hit a number of Ukrainian targets, there are also numerous misses. Even if it hits it does not guarantee a kill, sometimes causing minor damage that can be repaired. According to *Forbes*, Russia may be prevented from producing Lancets in large numbers due to international sanctions, as Russian drones are dependent on imported electronics and they are having difficulties in finding alternatives.^[17] However, in July 2023, television news footage showed a massive new facility with hundreds of the Lancets being manufactured as the makers claimed that production is being tripled.^[18]

According to Oryx, Dutch open-source intelligence website, as of 3 March 2023, Lancet drones scored more than 100 hits on Ukrainian targets during the war. Most of the targets were towed artillery and self-propelled artillery systems.^[19] A Russian source has also made claims about the high efficiency of the drone.^[20]

Since Spring 2023, Ukrainian forces have started to build cages around their artillery pieces using chain link fencing, wire mesh and even wooden logs as part of the construction. One analyst told Radio Liberty that such cages were “mainly intended to disrupt Russian Lancet munitions.” A picture supposedly taken from January 2023 shows the rear half of a Lancet drone that failed to detonate due to such cages. So the cages appear to work part of the time. Likewise Ukrainian forces have used inflatable and wooden decoys, shaped like vehicles and weapons systems, such as HIMARS, to confuse and divert Lancet attacks.^{[21][22]}

On 18 April 2023, reports emerged of the destruction of a Ukrainian patrol boat on the Dnipro River after being struck by a Lancet drone.^[23] On 27 April 2023 a video confirmed a Lancet strike on Ukrainian Tor SAM system.^[24] On 27 April, videos showing the destruction of one 5P85S launcher for the S-300 missile system and damaging of another by Lancet drones appeared on social networks.^[25]

During the 2023 Ukrainian counteroffensive, a TRML-4D multi-functional radar from the IRIS-T SLM air defense system was destroyed, as well as a Leopard 2A6 tank was heavily damaged by Lancet drones.^{[26][27]} On 22 July 2023, two Lancet drones destroyed a Spanish Aspide air defense system's fire control station and a launcher.^[28]

According to LostArmour open-source intelligence website, as of 3 August 2023, Lancet have conducted a total of 507 assaults on various Ukrainian weaponry since July 2022. Among these, 170 targets were subjected to destruction, while 269 armaments suffered substantial impairment. Lancet experienced

misalignment in 28 instances, and outcomes remain unconfirmed for 40 sorties. Over half of Lancet's engagements focused on artillery installations, encompassing 106 self-propelled cannons, 131 howitzers and mortars, and 18 multi-rocket launchers. Lancet destroyed or damaged 41 defense complexes of diverse kinds, along with 43 radar and communication hubs. Additionally, the drone caused destruction or impairment to 67 tanks.^[29]

Variants

Lancet-3

Basic and bigger variant with a 40-minute endurance with a maximum payload of three kilograms (6.6 lb) and a 12 kg (26 lb) MTOW. The maximum speed is 80–110 kilometres per hour (50–68 mph).^{[3][6]}

Lancet-1

A smaller version of the Lancet-3. It carries a one-kilogram (2.2 lb) payload and has a 5 kg (11 lb) MTOW. The endurance is 30 minutes.^{[3][6]}

Izdeliye-53 (or Product 53)

The latest version of the Lancet drone. The drones have a new tube launcher which can contain up to four Lancets and the drones can relay information between themselves about armored vehicles and their concentration and attack fully autonomously, choosing targets from pre-set categories. It has a maximum payload of five kilograms (11 lb).^{[30][31]}

Russian troops in Ukraine have begun using upgraded Lancet loitering munitions with an increased flight duration of one hour and a more powerful warhead weighing more than five kilograms, which is at least two kilograms more than the basic version of the drone. Lancets are used to attack personnel with a high explosive fragmentation or thermobaric warhead, and armored vehicles with a high-explosive anti-tank (HEAT) warhead.^{[32][33]} The upgraded variant is designated Izdeliye 51, while the original one, with a three-kilogram warhead, is designated Izdeliye 52 and reportedly was upgraded with a larger warhead and a new EO guidance system as of March 2023.^{[34][35]}

Operators

Russia

-  Russian Ground Forces

See also

- AeroVironment Switchblade
- UVision Hero
- HESA Shahed 136
- Artillery observer

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